ChatPRCS: A Personalized Support System for English Reading Comprehension Based on ChatGPT

Xizhe Wang, Yihua Zhong, Changqin Huang, *Member, IEEE*, and Xiaodi Huang, *Senior Member, IEEE*

***Abstract*—Reading comprehension is a widely adopted method for learning English, involving reading articles and answering related questions. However, the reading comprehension training typically focuses on the skill level required for a standardized learning stage, without considering the impact of individual differences in linguistic competence. This article presents a personalized support system for reading comprehension, named chat generative pretrained transformer (ChatGPT)-based personalized reading comprehension support (ChatPRCS), based on the zone of proximal development (ZPD) theory. It leverages the advanced capabilities of large language models, exemplified by ChatGPT. ChatPRCS employs methods, including skill prediction, question generation and automatic evaluation, to enhance reading comprehension instruction. First, a ZPD-based algorithm is developed to predict students' reading comprehension skills. This algorithm analyzes historical data to generate questions with appropriate difficulty. Second, a series of ChatGPT prompt patterns is proposed to address two key aspects of reading comprehension objectives: question generation, and automated evaluation. These patterns further improve the quality of generated questions. Finally, by integrating personalized skill prediction and reading comprehension prompt patterns, ChatPRCS is validated through a series of experiments. Empirical results demonstrate that it provides learners with high-quality reading comprehension questions that are broadly aligned with expert-crafted questions at a statistical level. Furthermore, this study investigates the effect of the system on learning achievement, learning motivation, and cognitive load, providing further evidence of its effectiveness in instructing English reading comprehension.**

***Index Terms*—Chat generative pretrained transformer (ChatGPT), personalized question generation (QG), prompt engineering, reading comprehension.**

Manuscript received 30 August 2023; revised 29 February 2024, 31 March 2024, and 21 April 2024; accepted 12 May 2024. Date of publication 27 May 2024; date of current version 13 June 2024. This work was supported in part by the National Science and Technology Major Project under Grant 2022ZD0117105, in part by the National Natural Science Foundation of China under Grant 62007031 and Grant 62337001, in part by the "Pioneer" and "Leading Goose" R&D Program of Zhejiang under Grant 2022C03106, and in part by Zhejiang Provincial Philosophy and Social Sciences Planning Project under Grant 24ZJQN083Y. *(Corresponding author: Changqin Huang.)*

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Zhejiang Normal University.

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Digital Object Identifier 10.1109/TLT.2024.3405747

I. Introduction

Problem-based learning (PBL) is a commonly used and effective instructional approach. Teachers typically design questions and tasks to facilitate instruction and evaluate students' understanding of the material. Reading comprehension is consistently incorporated into PBL as a common task in language education [1]. This task involves various question types and difficulty levels to evaluate learners' comprehension and information-processing skills. However, due to the complexity of question design and assessment development, teachers frequently depend on preprepared learning materials for instruction. This includes using standardized questions for the entire class during reading comprehension activities. Although this approach facilitates learning assessment, it has a limited impact on developing diverse levels of reading comprehension skills.

Currently, numerous studies have offered diverse theoretical guidance schemes aimed at enhancing learners' reading comprehension skills [2], [3]. During the actual cultivation process, akin to Bloom's Taxonomy, reading comprehension evolves through various stages: discovery, understanding, inference, induction, and expansion. Moreover, to enhance the cultivation of reading comprehension skills, beyond offering inquiry-based learning methods like PBL, two key issues must be addressed. 1) Accurate identification of learners' reading comprehension levels and alignment of developmental stage goals. 2) Provision of exploration-oriented guiding questions tailored to learners' objectives, aiming for tangible enhancements in reading comprehension skills.

The theory of the zone of proximal development (ZPD) [4] posits that tailoring content to match individual learners' ZPD fosters motivation, encourages learning, and facilitates reaching their potential and achieving higher levels of competence. Conversely, straying too far above or below learners' ZPD level hampers their skill development. Therefore, when instructing English reading comprehension, using questions at a standard level of difficulty for students with robust English reading skills, i.e., falling below their ZPD, can lead to an unengaging teaching process. In contrast, employing questions that are too challenging for students with relatively weak reading capabilities can result in frustration and a propensity to discontinue learning. In both scenarios, students' motivation and self-confidence are impacted [5], [6], leading to inefficient utilization of educational resources and a decline in educational quality. Therefore, offering personalized assistance to learners using artificial intelligence (AI) techniques has emerged as a crucial approach to enhancing the effectiveness of language learning. This assistance involves presenting questions customized to appropriate difficulty levels that align with the students' ZPD.

In recent years, AI has made significant strides in recognizing learners' states and generating personalized content. For example, chat generative pretrained transformer (ChatGPT) offers robust technical support for automatic question generation (QG), problem-solving, and tutoring reviews [7]. Existing studies suggest that the quality of generated questions has reached, and in some cases surpassed that of traditional teacher-crafted questions [8]. These generated questions not only effectively support learners' reading comprehension training and evaluation [9], but also alleviate the workload of teachers. However, the generation process does not consider students' current skill level, or the appropriate question difficulty levels that go with it [10]. Besides, existing algorithms primarily emphasize accurately predicting target results, but they often neglect to include a margin of error to align with learners' ZPD. Moreover, in terms of the representation of learning status, most studies typically focus on the instantaneous state of those features, disregarding potential changes induced by forgetting. Therefore, to achieve personalized support for reading comprehension learning, the following challenges must be addressed: First, personalized guidance must access data about individual learners' skills and be able to simulate the forgetting process through a time-varying function (i.e., forgetting curve). Second, an algorithm for measuring learners' current ZPD needs to be developed, enabling the provision of questions of appropriate difficulty—aligned with the current skill level or slightly advanced. Finally, the function of QG should be to respond to learner requests with precision, providing pertinent reading comprehension support without imposing additional cognitive load.

To address the aforementioned challenges, this study introduces the ChatGPT-based personalized reading comprehension support (ChatPRCS) system, to make the following three key contributions.

1) Propose a new personalized prediction algorithm for reading comprehension skills, based on the forgetting curve and the ZPD theory.
2) Design a series of ChatGPT prompt patterns tailored for English reading comprehension learning, including both QG and automatic evaluation patterns.
3) Implement an adaptive reading comprehension support system called ChatPRCS and conduct systematic experiments to verify its effectiveness.

The rest of this article is organized as follows. In Section II, the related literature is reviewed. Section III outlines the objectives and the questions to be addressed. Section IV presents an approach for predicting reading comprehension skills and designing the ChatGPT prompt patterns. Section V introduces the system interface and application mechanism of ChatPRCS. Section VI describes the participants, parameter settings, measuring tools, and procedures employed in the experiments. Sections VII and VIII report the experimental results and provide in-depth discussions. Finally, Section IX concludes this article.

II. Related Work

A. English Reading Comprehension Skill

The literature on factors affecting English reading comprehension can be classified primarily into categories of linguistic and nonlinguistic factors.

Linguistic factors refer to linguistic components, such as students' vocabulary knowledge and syntactic knowledge. Barnett [11] found a close correlation between the level of mastery of syntactic knowledge and reading comprehension. Similarly, Qian [12] showed that the level of vocabulary knowledge is one of the important factors affecting reading comprehension.

Nonlinguistic factors refer to factors unrelated to language itself, including previous reading performance, reading time, learning engagement, reading accuracy, and learning strategies. Previous learning performance refers to the experience learners have gained during their earlier learning endeavors. Varying learning performances play a significant role in affecting students' subsequent learning of English reading comprehension. Students with different levels of prior knowledge exhibit varying levels of reading quantity and quality [13]. Therefore, to help students comprehend English texts better, teachers should carry out differentiated teaching activities according to students' prior performance [14]. Besides, learning engagement is also one of the crucial factors affecting reading performance, which mainly includes cognitive, affective, and behavioral aspects [15]. Different levels of engagement will lead to varying learning outcomes [16]. Scholars also regarded reading time as an important factor in reading comprehension level [17]. However, only engaged reading time may contribute to students' reading performance positively. Studies suggested that teachers should motivate students for more effective time management to improve their effective engagement [18]. Furthermore, reading accuracy is another key factor in reading comprehension that is highly correlated with students' language development [19]. Students with poor reading accuracy may have to allocate most of their attention to extracting meaning from individual words [20]. Consequently, they often perform lower levels of comprehension skill (e.g., retelling, recall, etc.). In contrast, students with high reading accuracy can direct more attention to develop higher level comprehension skills (e.g., integration and reasoning) [21].

In summary, researchers have primarily focused on nonlinguistic factors to enhance learners' reading comprehension and have also attempt to design teaching methods that combine intelligent technology and effective strategies [22]. However, most of these approaches target the average level of the entire class with little consideration for the varying skills of individual learners. This is detrimental to the progress of both high- and low-ability students. Due to significant advancements in AI technology, particularly in the field of natural language processing (NLP) technology, more robust technical support for personalized learning is now available for both teachers and students [23], [24]. Therefore, this study aims to develop a system using AI technology to provide personalized learning support for learners according to their distinct reading comprehension levels.

B. AI-Empowered Personalized Systems for English Learning

AI technology can provide personalized guidance, support, and feedback to students while also aiding teachers in their teaching process [25]. This claim is supported by a large body of research, where scholars have developed personalized learning systems by leveraging machine learning algorithms and data analytics techniques [26]. These systems are designed to provide individualized support for learners of different skill levels. Addressing the challenge of the absence of authentic context in language learning [27], Chen et al. [28] developed a language learning system. The system incorporates AI and GPS functionalities to facilitate students' contextualized English learning. The experimental results showed that the use of the system could enhance students' motivation and academic performance.

In recent years, many researchers have progressively turned to the utilization of generative artificial intelligence (GAI) technologies to offer support for teachers and students [29], [30]. For example, Chomphooyod et al. [31] developed an automated generation system based on GAI, capable of automatically generating questions for students and aiding teachers in the evaluation. This system can improve students' performance and reduce teachers' burden. Within the realm of language learning, researchers have utilized ChatGPT's extensive language knowledge in students' writing endeavors [32]. By providing timely feedback [33], ChatGPT has effectively improved students' motivation and learning performance. Moreover, GAI-based dialogue systems [34] and robots [35] have found application across various aspects of English learning, such as reading and speaking, and have achieved good results.

The aforementioned studies have demonstrated the capacity of GAI to enhance students' personalized learning. However, they still have limitations. For instance, they neglect to incorporate students' skill levels for personalized QG [31], resulting in the generation of questions misaligned with students' current skill levels. Furthermore, most of the existing studies concentrate solely on one factor of reading comprehension when making predictions about reading skills [1]. This approach largely overlooks other potential factors influencing students' reading performance. By using state-of-the-art AI tools, this study predicts learners' reading skills by considering multiple factors that affect their reading comprehension learning within a system. As such, the system can achieve improved accuracy in providing personalized learning support.

III. Objectives and Research Questions

In this study, the ChatPRCS system was developed to offer personalized support for enhancing English reading comprehension instruction. To address the abovementioned challenges that remain in the realm of fostering English reading comprehension skills, the ChatPRCS system has been devised to answer the following questions.

RQ1. How to quantify students' reading comprehension skills using the forgetting curve and ZPD theory to facilitate personalized QG?

RQ2. What about the quality of the generated questions given by the ChatPRCS system? Can the generated questions meet students' personalized needs for reading comprehension?

RQ3. Does the use of the ChatPRCS system for English reading comprehension learning result in increased levels of students' learning achievement, learning motivation, and cognitive load?

IV. Methods: ZPD-Based Predictions for Reading Comprehension Skill and Prompt Patterns

As illustrated in Fig. 1, the ChatPRCS system consists of three primary modules: input, learning activities, and data collection. The input module encompasses learning content (input ①, uploaded by teachers), reading comprehension skills (input ②), and a repository of historically generated questions (input ③). For learning activities, the system has two functions for reading comprehension: Personalized QG, and automatic evaluation. These capabilities are underpinned by the ChatGPT framework. In addition, a dedicated prompt pattern has been devised for each function, offering students the option to select the pattern that improves the effectiveness of the generated results. Finally, the data collection module gathers and updates data on students' reading comprehension skills. These data inform subsequent rounds of personalized QG, thereby providing more effective assistance to students in their learning activities.

A. Quantifying Reading Comprehension Features

Assessing English reading comprehension skills encompasses a comprehensive evaluation of vocabulary, comprehension, and content organization. Many previous studies have indicated a positive correlation between learners' reading skills and factors, such as knowledge level [1], learning performance [13], and learning duration [36]. In alignment with the correlations identified in existing research on English reading comprehension skills [19], [37], and leveraging the data that can be collected by the ChatPRCS system, this study integrates four features to represent an individual learner's English reading comprehension skill G: Academic knowledge level K, effective learning duration D, historical performance H, and answer precision P. For the ith learner providing answers to the kth question during the jth practice session, each feature is quantified and characterized as detailed in the following.

Academic knowledge level K: Similar to a learner's academic grade, knowledge growth follows an extended path that typically remains stable throughout a course. Therefore, this feature was considered a static value for each learner within this study. Based on the system's records of each learner's knowledge and skill level in English content (self-evaluation), with reference to the Public English Test System (PETS) scale [38], the ith learner was assigned a score on a scale of 1 to 5, i.e., $K^{(i)} \in \{1, 2, 3, 4, 5\}$.

Effective learning duration D: Generally, the time taken to answer questions reflects learners' familiarity with vocabulary and article content [17]. Due to the fragmentation and uncertainty of online learning environments, recording the effective time for answering questions can more accurately reflect students' reading comprehension abilities. Therefore, the ChatPRCS system incorporates an anchor point detector to identify the specific

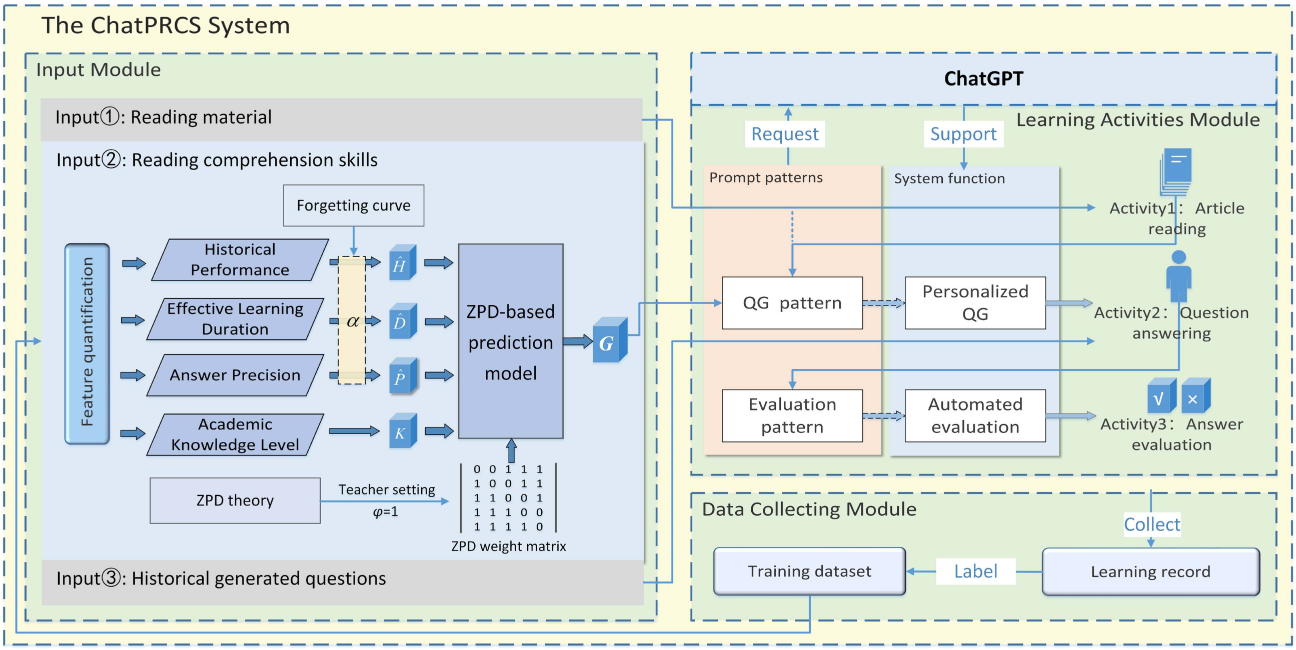

Fig. 1. Structure of the ChatPRCS system.

operations (such as window sliding and mouse action in the system) performed during the answering process. Suppose a learner conducts a total of the V operations within an answering process, and $T_v^{(i)}$ is the system-recorded time of each operation for the ith student performing the vth operation. By setting an activity threshold denoted as λ, valid behavioral records can be identified. Consequently, effective learning time is computed by calculating the time for each operation as follows:

$$D^{(i)} = \sum_v T_v^{(i)}, T_v^{(i)} = \begin{cases} T_v^{(i)}, & T_v^{(i)} < \lambda \\ \lambda, & T_v^{(i)} \geq \lambda. \end{cases} \quad (1)$$

Historical performance H: Historical performance represents the rating of answers to questions from a previous period. The score of historical answers (especially for similar theme reading material) is another key factor in measuring learners' reading skills. Learners' responses are automatically recorded by the ChatPRCS system, which can be used to measure learners' English reading comprehension skills. For the ith learner's responses to answer the kth specific question $\mathrm{AC}_k^{(i)}$, based on the evaluation pattern EP(), the historical performance is given by

$$H_k^{(i)} = \mathrm{EP}(\mathrm{AC}_k^{(i)}). \quad (2)$$

Answer precision P: Generally, the criteria for evaluating an essay question are mainly based on whether the answers include all the key points of the knowledge to be tested. In other words, if an answer contains all the key points, the student who provides this answer will very likely receive a high mark, even if it includes much irrelevant information. However, in addition to identifying all key information, reading comprehension tasks often further examine the learner's ability to summarize key points. A concise and comprehensive expression also reflects a higher level of reading comprehension. Therefore, the precision of an answer was calculated according to the hit rate using the semantic similarity of keywords. For the nonstop word text sequences provided by learners answering the total n questions in jth exercise, denoted as $\{\mathrm{ac}_1, \ldots, \mathrm{ac}_p\}$, and the corresponding nonstop word text sequence of the correct answer $\{\mathrm{rc}_1, \ldots, \mathrm{rc}_q\}$. The precision of the answer was calculated as follows:

$$P_j^{(i)} = \frac{\sum_n \sum_p \sum_q \mathrm{sim}\left(\mathrm{ac}_p, \mathrm{rc}_q\right)}{p+q}. \quad (3)$$

In (3), p represents the total number of keywords answered by the student, and q is the total number of keywords in the answer. The function $\mathrm{sim}(c_1, c_2)$ calculates the semantic distance between texts c_1 and c_2 based on our previous studies [39], [40].

Reading comprehension skill G: As the output of the prediction model, reading comprehension skill is categorized into five levels, denoted as $G^{(i)} \in \{1, 2, 3, 4, 5\}$. Initially, these skill labels were assigned by teachers, where 1 represents the lowest level and 5 represents the highest level. In addition, across the entire sample, the distribution of these skill levels approximates a normal distribution. This value serves as a basis for determining question difficulty when generating questions.

The cognitive abilities of learners evolve over time due to forgetting, resulting in fluctuating values of features when predicting reading comprehension skills. As such, the forgetting curve is used to delineate the shifts in cognitive patterns as time elapses. Drawing inspiration from Ebbinghaus's law [41] and aligning with the intricacies of the reading comprehension task, a temporal influence threshold is denoted as a, based on the specific jth session. This threshold is derived as follows:

$$a_j = e^{-\frac{r(t-t_0)}{j+1}}. \quad (4)$$

In (4), r is denoted as the parameter of knowledge retention in the reading comprehension task, t means the current time, and t_0

is the most recent time of satisfactory memory retention. These parameters collectively form the forgetting weights specifically tailored to the reading task.

To better describe the impact of forgetting, the input of the prediction model is formed by multiplying each affected feature with its corresponding forgetting weight, yielding $\hat{X}^{(i)} = a_j X^{(i)}$. Consequently, a set of predictive features is formed by considering the influence of forgetting effects

$$\hat{X} = \left\{K, \hat{D}, \hat{H}, \hat{P}\right\}. \tag{5}$$

In line with the intricacy of reading comprehension problems, a prior study [42] has preliminarily characterized learners' reading comprehension skill as $\sum(\text{feature}_N \times \text{weight}_N)$. Through training on available data, the model can learn the weights of various features, enabling it to predict reading comprehension skills. However, this method lacks consideration for task-specific adaptability with low accuracy. Therefore, it is necessary to design an effective capability prediction scheme based on the characteristics of the prediction task.

B. Predicting Reading Comprehension Skill

According to the ZPD theory [4], questions that are excessively challenging or overly simple have minimal influence on learners' skills development. Instead, providing learners with questions that align with their skills, or slightly surpass them, can foster effective learning [43]. However, traditional instructional approaches typically determine question difficulty based on a general assessment of learners' skills and prior performance. This approach may overlook individual differences and hinder the development of each student's unique strengths. Consequently, to offer personalized reading comprehension questions suitable for diverse learners, it is essential to develop a prediction method capable of assessing each learner's skill in English reading comprehension. The outcome of this prediction would then serve as a basis for subsequent QG.

In this study, the skill prediction model aims to provide learners with results that are consistent with their current skill level or slightly higher, to maximize skill improvement. On the contrary, if the predicted outcomes fall notably below or above the actual skill level, this could hinder the enhancement of reading comprehension skills. Therefore, inspired by MetaCost [44], this study proposes a skill prediction model rooted in the ZPD theory. Through experiments with various multiclassification machine learning models, e.g., multiple discriminant analysis (MDA) [45], multinomial logistic regression (MLR) [46], decision tree (DT) [47], neural networks (NN) [48], the optimal solution for predicting the current reading comprehension level of students is determined. The formalization of the task description and ZPD weight matrix are given as follows.

Definition 1 (Task description): Given the training set $D = \{\hat{X}, G\}$ consisting of M samples, each characterized by N distinct features, these samples are classified into L classes, denoted as $\hat{X} \in \mathbb{R}^{M\times N}$, $G \in \mathbb{R}^{L\times M}$. To achieve this, one-hot coding is used to label each sample in G. This means that in the vector $G^{(i)}$, only one element equals 1 while the others are 0.

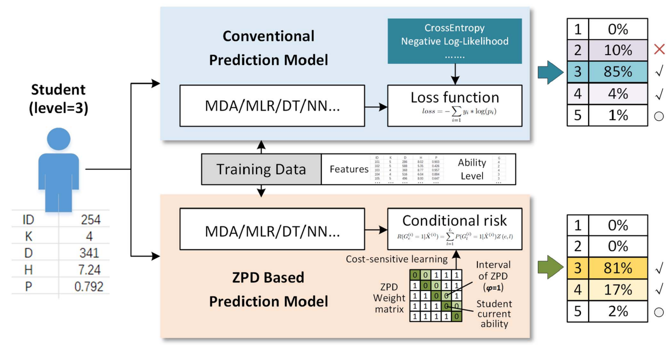

Fig. 2. Example of use cases for the difference between the conventional prediction model and the ZPD-based prediction model.

For any given sample and its corresponding label, the probability that the predicted result c belongs to the correct category l can be denoted as

$$P\left(G_c^{(i)} = 1|\hat{X}^{(i)}\right). \tag{6}$$

Consistent with the ZPD goal, it is expected that the predicted results will align with or slightly exceed the learner's current level. For example, in Fig. 2, if a learner's current reading skill level=3, the prediction result is most likely to be 3 (with the highest probability among the five levels), or it may be 2 or 4 without using ZPD. This depends on the different importance of categories when using ZPD. In other words, although models with ZPD still have the highest probability of suggesting he/she at the level of $G = 3$, there is also a relatively higher possibility of suggesting $G = 4$ instead of suggesting $G = 2$. In order to differentiate the varying importance of each category, different weights for the category result c from the model are calculated as follows.

Definition 2 (ZPD weight matrix): $Z \in \mathbb{R}^{L\times L}$ is defined as the weight matrix representing different categories in predicted results. Here, $Z(c, l)$ means the weight assigned to classifying an instance into category c. When $c = l$, we have $Z(c, l) = 0$. For a predicted result category $l(l < L)$, given a ZPD interval threshold φ (set to 1 by default), this value can be modified by the teacher based on the difficulty of the article and the learning status of the students. The ZPD weight matrix of the upward-floating category is calculated in the following equation:

$$R\left(G_c^{(i)} = 1|\hat{X}^{(i)}\right) = \sum_{l=1}^{L} P\left(G_l^{(i)} = 1|\hat{X}^{(i)}\right) Z(c, l). \tag{7}$$

By minimizing R as much as possible, the classification results of the model are more likely to fall within the ZPD range. As such, it ensures that the predicted results meet the task requirements. Thus, the model training algorithm is detailed in Algorithm 1.

Fig. 2 illustrates an example of use cases highlighting the disparity between the conventional prediction model and the ZPD-based prediction model. While the conventional prediction model demonstrates a higher probability (85% compared to

Algorithm 1: ZPD-based Prediction Model Training

Input: training set $D = \{\hat{X}, G\}$, base classifiers $\mathcal{M}$, weight matrix Z, number of resamples to generate N_{rg}, and number of examples in each resample N_{er}.
Output: The final classification model $\mathcal{M}'$.

for $s = 1$ to N_{rg} **do**
  Let D_s be a resample of D with N_{er} examples;
  Let $\mathcal{M}_s$ be the model produced by applying $\mathcal{M}$ to D_s;
end for
for each example $\hat{X}^{(i)}$ in D **do**
  for each class l **do**
    Calculate $\frac{1}{N_{rg}} \sum_s P(G_l^{(i)} = 1|\hat{X}^{(i)},\ \mathcal{M}_s)$;
  end for
  Calculate Z for the task by Eq. (7);
  Calculate R of example $\hat{X}^{(i)}$ by Eq. (8);
  Relabel the class of $\hat{X}^{(i)}$ as $\text{argmin}_s R(s|\hat{X}^{(i)})$;
end for
Let $\mathcal{M}' =$ Model produced by applying $\mathcal{M}$ to D;
return $\mathcal{M}'$.

81%) of accurate predictions at level 3, there is also a notable chance of a level 2 prediction (10%). However, employing level 2 difficulty as the criterion for generating subsequent problems fails to foster learner skills. The prediction model, based on the ZPD and operating under the cost-sensitive learning paradigm, may not achieve the same prediction accuracy as conventional models. However, it can ensure that the predicted results are likely to fall within the range of levels 3 to 4, accounting for a total of 99%. This result is more in line with the needs of learners in the ZPD.

C. ChatGPT Prompt Patterns for English Reading Comprehension Learning

The key advantage of ChatGPT lies in its user-friendly interface combined with its powerful text generation capabilities. Users simply specify their requirements by typing text into the input box, much like sending a message from a terminal. After clicking the "generate" button, ChatGPT processes the input to produce results of expert quality. These inputs could range from inquiries like "What is the definition of artificial intelligence?" to requests for background information, such as "Tell me about the planets in the solar system." or even seeking assistance as in "Could you assist me writing a brief introduction about artificial intelligence?" The nature of the response received depends on the user's input. Thus, effective communication with conversational large language models (LLMs) such as ChatGPT is crucial to cater to the users' requirements.

Prompt engineering serves as a programming approach that enables users to tailor the output and interaction with LLMs [49]. By giving instructions to LLMs, users can impose rules, automate processes, and ensure the generation of output with specific quantified quality and volume. Given the widespread use of LLMs in various domains, prompt engineering has evolved into an increasingly indispensable skill set. As such, researchers

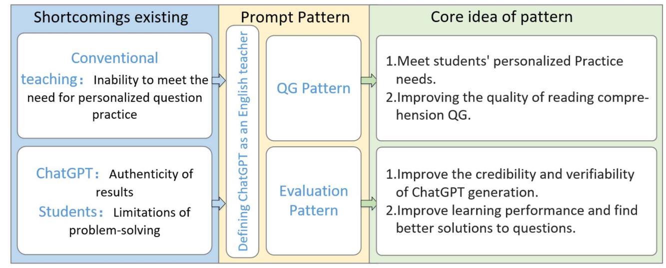

Fig. 3. Introduction of prompt patterns.

have embarked on diverse investigations, such as the "prompt pattern catalog" proposed by White [50]. This catalog collects and organizes the prompt patterns of pretrained models across various NLP tasks. It provides a rich and customizable repository of prompt templates, facilitating users to better utilize the pretrained models for NLP tasks.

The proposed approach for the quantification and prediction of students' reading comprehension skills and the final G obtained is described in Sections IV-A and IV-B. This enabled ChatGPT to discern the student's English reading comprehension skill level and use it as a parameter for the difficulty levels of QG. This provides the foundation for the subsequent system components focused on personalized generation. After receiving the student's information, ChatGPT was enabled to retain the student's current data and offer the relevant English learning materials tailored to formal learning needs.

The problems to be addressed by the prompt pattern, as well as its core ideas, are described in Fig. 3. In these prompt patterns, the objective was that ChatGPT emulate an English teacher's role and provide expert-level guidance to students. To achieve this, ChatGPT takes on the role of a teacher first, ensuring that it is aware of its responsibility to function as a teacher when generating content. With this guidance in place, two specific prompt patterns were refined to align with the specific requirements of the system: one for QG and another for evaluation purposes.

1) QG Pattern: After reading the provided materials, students were required to answer the questions. Questions are a fundamental part of teachers' instructional methodology, serving various purposes including assessing students' skills and acquiring feedback on the effectiveness of teaching approaches. The design of questions requires thoughtful consideration of several factors, such as content relevance, alignment with students' skills, and incorporation of a variety of question formats (e.g., multiple choice, fill-in-the-blank, short answer, etc.). Table I shows an example implementation of the QG pattern.

Although the student's initial reading comprehension skill is assigned during the phase of the student information setup, this skill may evolve throughout the learning process. Therefore, instruction QG1, which states "Please keep in mind that the student's reading comprehension skill is G," inputs the student's skills into ChatGPT. This enables updates that align with the generation of questions tailored to their current reading comprehension level. Similarly, instruction QG2, which instructs "Please generate questions that align with the student's reading comprehension skill based on the articles provided," guides the

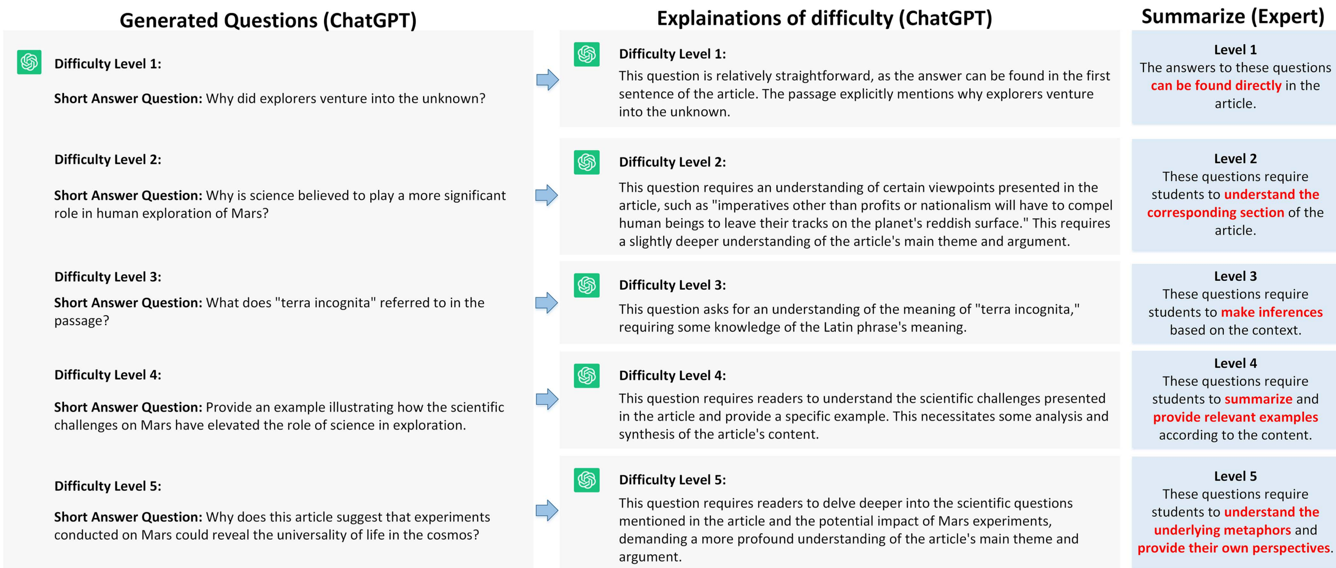

Fig. 4. Example of generated questions with different difficulty levels.

TABLE I
EXAMPLE OF QG PATTERN IMPLEMENTATION

QG prompt instructions (Abbreviated as Instruction QG)
1. Please keep in mind that the student's reading comprehension skill is G.
2. Please generate questions that align with the student's reading comprehension skills based on the articles provided. (tip: you can even specify the type and number of questions you would like to generate.)
3. Whenever I request a particular question type, feel free to generate a question that you believe to be a better one.
4. Please explain why you consider this question is better than the original one and ask my opinion.
5. Regenerated into a question with improved quality.

TABLE II
EXAMPLE OF EVALUATION PATTERN IMPLEMENTATION

Evaluation prompt instructions (Abbreviated as Instruction E)
1. Please assess the answers provided by the students.
2. Please provide the rationale underlying your evaluation results.
3. Please offer precise suggestions for rectification.

generation of questions based on the student's skill level and the provided articles. The case of generating questions with varying degrees of difficulty is illustrated in Fig. 4.

Instruction QG3, "Whenever I request a particular question type, feel free to generate a question that you believe to be a better one," mainly considers two aspects: 1) if the type of question specified by the user is not suitable, like preferring multiple choice questions over fill-in-the-blank or short-answer questions and 2) when asked to generate a higher quality question, ChatGPT naturally optimizes the question to be more suitable for assessing students.

Instruction QG4, "Please explain why you consider this question is better than the original one and ask my opinion" is grounded in two principles: 1) the user's potential challenge in distinguishing the most fitting practice question necessitates an explanation and 2) the user might be dissatisfied with the previously generated questions. In such a case, she may prefer to use instruction QG5, "Regenerated into a question with improved quality" to generate alternative questions.

2) Evaluation Pattern: In traditional teaching, the task of teacher evaluation is often time-intensive and subjective, leading to untimely and impersonal feedback. Automatic evaluation technology can better mitigate these challenges for both educators and students, thus improving the efficiency and quality of teaching and learning. However, automatic evaluation technology has drawbacks, such as the challenge of consistently generating realistic and convincing answers, which can make it difficult for users to discern their authenticity. Therefore, this study introduces an evaluation pattern to mitigate these deficiencies in automatic evaluation, thereby enhancing its accuracy. Table II shows an example implementation of the evaluation pattern.

After learners have answered the questions, the results are input into ChatGPT, and instruction E1, "Please assess the answers provided by the students" guides ChatGPT to automatically evaluate the results. To prevent misleading users from convincing but untrue answers generated by ChatGPT, instruction E2, "Please provide the rationale underlying your evaluation results," improves the credibility of these responses by prompting ChatGPT to elaborate on the specific justifications for its evaluation results.

In addition, students are often influenced by conventional and cognitive biases, which causes them to persist with a particular approach to problem-solving even after recognizing an error. This behavioral pattern requires using instruction E3, "Please offer precise suggestions for rectification," after evaluating students' answers. It empowers ChatGPT to suggest problem-solving adjustments and compare them against students' answers to determine their suitability. By doing so, this approach

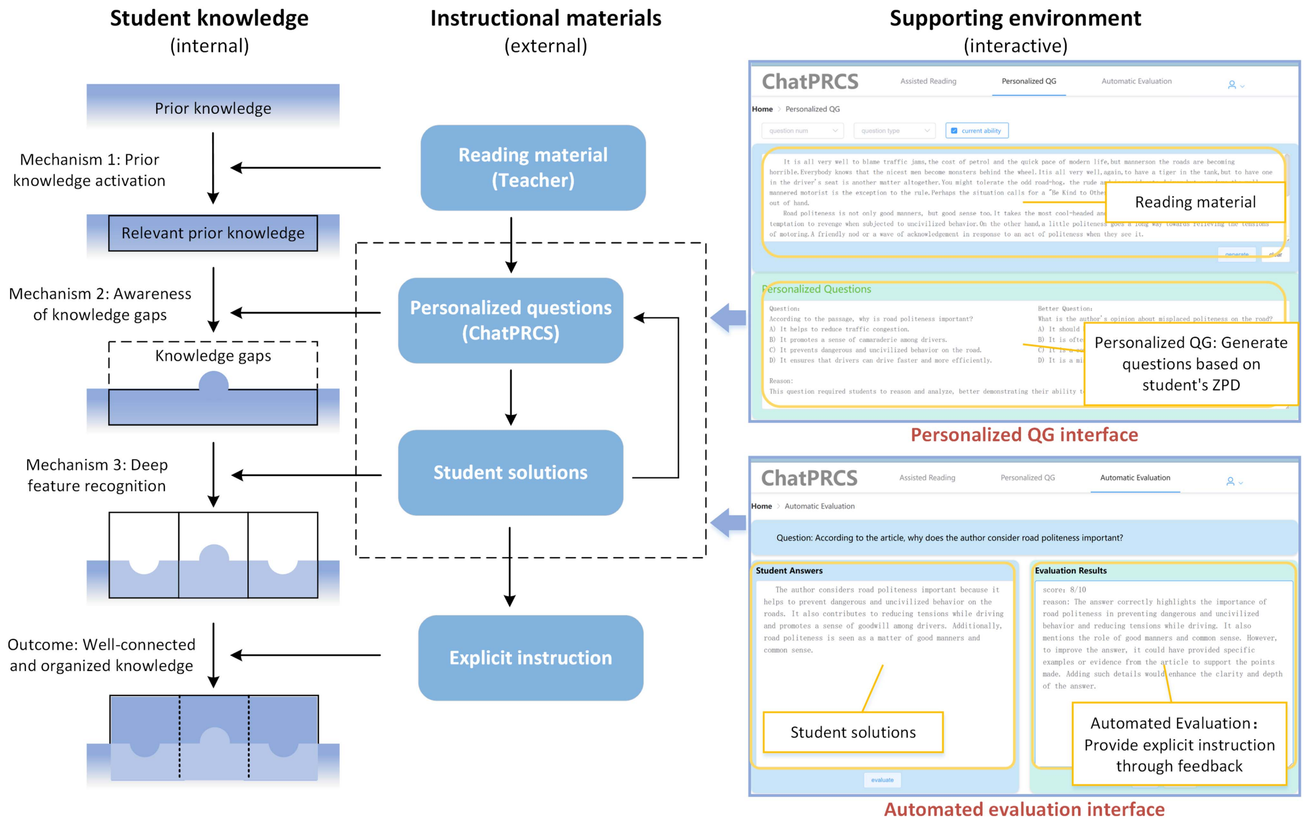

Fig. 5. Assisted learning mechanism and interfaces of ChatPRCS.

broadens students' cognition and reduces their cognitive bias, enabling them to find better solutions to problems.

V. Implementation and Learning Mechanism of ChatPRCS

A. Implementation of ChatPRCS

ChatGPT adopts a conversational structure for generating content, which streamlines the process of obtaining desired results for users. However, this study focuses on enhancing students' English reading comprehension, which does not naturally align with a dialogue format. Therefore, the ChatPRCS system was implemented to interact with ChatGPT by converting the users' operations on the client side into prompted instructions (prompted instructions outlined in Section IV-C). This approach not only simplifies the user's operational steps but also limits instances of misuse (such as generating answers alongside questions for cheating during exams). The system provides two primary functions: personalized QG and automated evaluation.

1) Personalized QG Function: The first step in the process of personalized QG involves obtaining the current skill level of students' reading comprehension skills, as depicted in Fig. 5. While student skills were initially configured as described in Section IV-B, their dynamic nature necessitates periodic updates in ChatPRCS. In addition, users can choose the number or types of questions in line with their preferences. ChatGPT generates questions based on the given article and shows the generated content in the system. This outcome is attributed to the prompt pattern introduced in this study, evident in QG3 within "The QG Pattern" detailed in Section IV-C.

2) Automated Evaluation Function: Faced with the dilemma of applying conventional evaluation methods to personalized questions, this study developed an automated evaluation function. As illustrated in Fig. 5, this interface displays the results of students' answers alongside the evaluated results provided by ChatGPT. ChatGPT offers rationales for evaluation, additional scores, error explanations, and revision suggestions based on "the evaluation pattern," denoted as E2 and E3. This functionality provides students with timely, personalized feedback, helping them correct errors and fostering a more adept master of the subject matter.

B. Assisted Learning Mechanism Based on ChatPRCS

Question tests serve not only as assessment tools but also as effective aids for student learning [51]. This study aims to utilize ChatPRCS not solely as an assessment tool but also as a learning aid. The personalized questions produced by ChatPRCS aim to stimulate students' critical thinking and enhance their reading comprehension skills, extending beyond merely detecting prior knowledge.

Building upon Loibl's theoretical framework [52], a learning mechanism was designed centered around ChatPRCS. As a supportive environment, ChatPRCS provides students with external resources, fostering various learning mechanisms including the activation of prior knowledge, awareness of knowledge gaps,

TABLE III
PARTICIPANT BACKGROUND INFORMATION

	Full sample	EG	CG
N	42	22	20
Male	18	9	9
Female	24	13	11
Average age	19.43 (SD=0.966)	19.32 (SD=0.839)	19.55 (SD=1.099)

Experimental group(EG), control group(CG).

and recognition of deep features throughout the learning process. The primary objective is to assist students in constructing internal knowledge and enhancing their competence, extending beyond the provision of personalized questions alone, as depicted in Fig. 5.

VI. EXPERIMENTAL DESIGN AND SETUP

The experiment was conducted in an English course at a Chinese university. The experiments of this study consist of two parts: the training of the prediction model using historical learning data and the validation of practical application using the ChatPRCS system.

A. Participants

To train and test the effectiveness of the model in selecting an appropriate prediction scheme, three English course teachers participated in labeling data based on historical data to categorize levels of reading comprehension skills. Through consistency tests, a dataset with a total of 295 training instances was used for model training and validation.

After that, a quasi-experimental study was conducted for four weeks to verify the effectiveness of the ChatPRCS system in the context of English reading comprehension learning. A total of 42 sophomore students (24 females and 18 males) majoring in educational technology participated in the experiment, and all participants provided informed consent forms. Their background information is given in Table III. These participants were distributed randomly into two groups: the first group was a control group with 20 students who followed a conventional approach using standardized questions for learning and practice. The second group consisted of 22 students who engaged in learning and practicing personalized questions through the ChatPRCS system.

B. Base Models and Parameter Settings

Selecting the optimal model for this task involves employing four multiclass classifiers: MDA [45], MLR [46], DT [47], and NN [48]. The stratified tenfold cross-validation was used in this experiment.

All experiments were performed on an RTX 3080 Ti GPU with Python 3.7 and sklearn 1.1.1. The initialization parameters including learning rate and weight were set to the default values in sklearn. Further parameters are listed in Table IV.

TABLE IV
PARAMETER SETUP

Parameter	Notes	Values
λ	Threshold of interaction duration	30 s
r	Parameter of knowledge retention level	0.85
φ	Interval of ZPD	1

C. Evaluation Metrics

Three different metrics are employed for comparing model performance: Accuracy, area under curve (AUC), and conditional risk R.

Classification accuracy is defined as follows:

$$\text{Accuracy} = \frac{\text{TP+TN}}{\text{TP+FP+TN+FN}}. \tag{8}$$

AUC is defined as

$$\text{AUC} = \frac{\text{sensitivity} + \text{specificity}}{2}. \tag{9}$$

R is calculated by (8).

In this study, the research instruments included pretest, posttest, questionnaires, and qualitative interviews. The investigation aimed to explore students' learning achievements, learning motivation, cognitive load, and the quality of machine-generated questions. The pretest and posttest questions were developed by two experts with extensive teaching experience. The questionnaires integrated a combination of machine-generated and expert-written questions, with each questionnaire consisting of ten questions, each assigned ten points, resulting in a total possible score of 100 points. The Cronbach's α values for the pretest and posttest were 0.79 and 0.89, respectively, indicating the questionnaires' reliability.

Regarding students' learning achievement, pretest and posttest data were analyzed to determine the learning accomplishments of both groups. Questionnaires were mainly used to investigate students' learning motivation and cognitive load. The learning motivation questionnaire was modified from the scale developed by Pintrich [53], which consists of six items on a 5-point Likert scale with a Cronbach's α value of 0.83. The cognitive load questionnaire was adapted from the scale established by Hwang et al. [54]. It comprises five items gauging mental load and three items measuring mental effort on a 5-point Likert scale, with Cronbach's α values of 0.77.

The quality of machine-generated questions was assessed through two primary dimensions: their alignment with expert-written questions and their difficulty level tailored to the students' current reading comprehension skills. Because of the black-box nature of ChatGPT, this study drew on the idea of the Turing test to assess these two aspects.

Regarding the first dimension, students were asked to determine whether the ten posttest questions were machine-generated or expert-written. Among these questions, the five machine-generated questions were specifically labeled Q2, Q3, Q6, Q7, and Q9. The students' skills to accurately discern the origin of these questions were evaluated. To enhance the reliability of the findings, different experts were also invited to rate the quality of the ten questions.

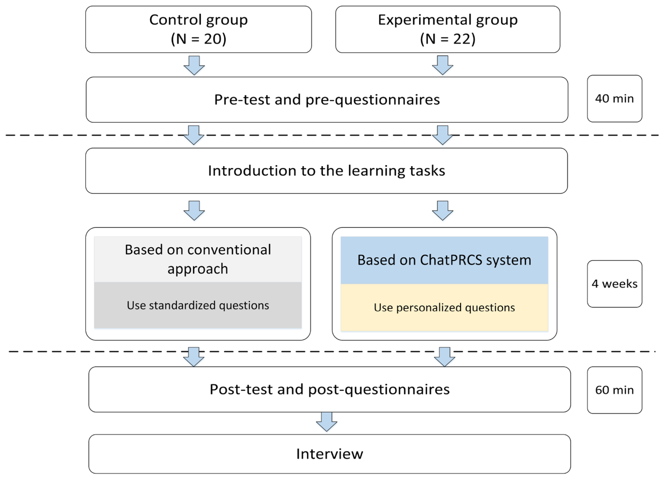

Fig. 6. Experiment procedure.

Addressing the second dimension, a survey was conducted to gauge the suitability of question difficulty. This survey has two question sets: machine-generated standardized questions (these questions were provided for the control group, and ChatGPT was generated without using the prompt pattern) and personalized questions (these questions were provided for the experimental group, and ChatGPT was generated using the QG pattern presented in Section IV-C), with each set containing tenn questions. The questions in the control group were labeled as CQ1–CQ10, while those in the experimental group were labeled as EQ1–EQ10. It should be noted that although the questions in the experimental group were all denoted with the EQ to indicate their personalized questions, they were distinct from each other. Finally, these questions were disseminated among both the control and experimental groups to assess whether the question difficulty corresponded to their current English reading comprehension skills. This evaluation was performed on a scale of 1–5 (1—Too simple, 2—Slightly simple, 3—Suitable, 4—Slightly difficult, and 5—Too difficult).

D. Experimental Procedure

The experiment was divided into three phases, with the specific experimental procedure outlined in Fig. 6. During the first phase, both the control and experimental groups took a pretest to evaluate English reading comprehension before starting the experiment. Participants completed a learning motivation questionnaire as well. During the second phase, the two groups participated in eight rounds of English reading comprehension learning. The control group used a conventional learning approach, which means using standardized questions. In contrast, the experimental group employed the ChatPRCS system to use personalized questions. For the third phase, this study collected and analyzed the data from the posttest, learning motivations, and cognitive load questionnaires from two groups. The results of the pre and posttests facilitated the following three core objectives:

1) evaluate whether students using personalized questions performed better than those using standardized questions in terms of learning achievement;
2) appraise the quality of the machine-generated questions;
3) explore students' learning motivation and cognitive load when using the ChatPRCS system for English reading comprehension learning.

Furthermore, an interview was also conducted to collect students' feelings and opinions in terms of utilizing the ChatPRCS system.

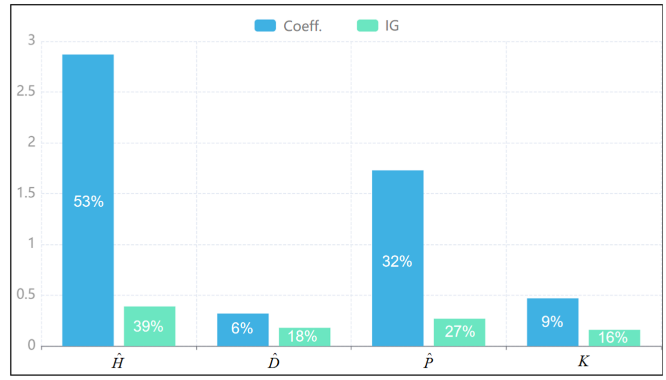

Fig. 7. Coefficients and IGs of reading comprehension features.

TABLE V
CLASSIFICATION PERFORMANCE

Classifier	Accuracy	AUC	R
MDA	71.36	0.731	7.34
MLR	81.23	**0.823**	**6.92**
DT	82.31	0.817	8.63
NN	**84.96**	0.763	9.17

MDA: multiple discriminant analysis, MLT: multinomial logistic regression, DT: decision tree, NN: neural networks
The best results for each metric (row) are bolded.

VII. EXPERIMENTAL RESULTS AND ANALYSIS

A. How to Quantify Students' Reading Comprehension Skill Using the Forgetting Curve and ZPD Theory to Facilitate Personalized QG?

1) Feature Importance Ranking: The model prediction results were presented from two perspectives. To evaluate the impact of features, the MLR and DT classification processes were selected from the four models by using the coefficient (coeff.) of the features and the information gain (IG) in node splitting. Both of these values can be used as discriminative indicators of feature importance, as shown in Fig. 7.

In the presented results, despite notable differences in specific values between the two indicators, there exists a certain degree of consistency in their overall proportion distribution. Notably, $\hat{H}$ and $\hat{P}$ exhibit higher scores overall, with $\hat{H}$ particularly excelling on both indicators. Conversely, $\hat{D}$ and K obtain relatively lower scores on both indicators. This observation underscores the predominant role played by the historical answer performance feature in accurately predicting reading comprehension skills.

2) Model Performance Comparisons: The results of model prediction based on the three evaluation metrics are reported in Table V. In terms of accuracy, NN achieved the highest accuracy,

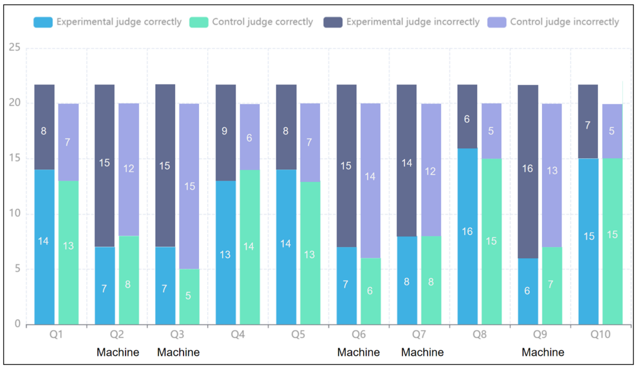

Fig. 8. Number of students who judged the questions correctly and incorrectly.

but it also yielded the highest R value. DT and MLR demonstrated comparable results in accuracy and AUC. However, when considering the result of R, MLR has an advantage. Compared to the other models, MDA performed poorly in both accuracy and AUC, indicating its unsuitability for this task. Given these considerations, and acknowledging that MLR achieved superior overall prediction accuracy, AUC, and the lowest R value, which can be selected as the optimal method for predicting learners' reading comprehension skills in the ChatPRCS system.

B. What is the Quality of the Generated Questions by the ChatPRCS System? Can the Generated Questions Meet Students' Personalized Needs for Reading Comprehension?

1) Assessing the Quality of Two Type Questions: Fig. 8 shows the results of students' judgments on the two question types. Among the five machine-generated questions, namely, Q2, Q3, Q6, Q7, and Q9, a total of 7 (32%), 7 (32%), 7 (32%), 8 (36%), and 6 (27%) students in the experimental group, respectively, judged correctly. Similarly, 8 (40%), 5 (25%), 6 (30%), 8 (40%), and 7 (35%) students in the control group, respectively, judged these questions correctly. For the five expert-written questions, specifically Q1, Q4, Q5, Q8, and Q10, 14 (64%), 13 (60%), 14 (64%), 16 (73%), and 15 (68%) students in the experimental group made correct judgments. In the control group, correct judgments were made by 13 (65%), 14 (70%), 13 (65%), 15 (75%), and 15 (75%) students. From these data, it was clear that both the experimental and control groups exhibited a notably higher likelihood of accurate judgments when presented with expert-written questions compared to machine-generated questions. Upon exploring the rationale behind this divergence, during interviews with the students, many reported that the perceived quality of the questions was quite comparable. It was difficult for them to judge these questions apart, so they often resorted to guessing when faced with the task of distinguishing between the two types of questions. Subconsciously, they believed that the machine-generated questions were unlikely to attain the same level of quality, so they all tended to categorize such questions as being expert-written.

To enhance the conclusions' reliability, three experts were invited to rate the question quality, using a scoring scale ranging from 0 to 100. Table VI presents the results of descriptive statistics and independent sample t-tests conducted on the expert ratings. The average scores assigned by the three experts to the machine-generated and expert-written questions were 82.73 and 81.66, respectively. This result shows no significant difference in the quality of the two question types (t = 0.47, p >0.05). This implies no statistically significant difference in the quality between machine-generated and expert-written questions.

TABLE VI
RESULTS OF THE t-TEST ON EXPERT RATING

Variable	N	Mean	SD	t	d
Machine-generated questions	5	82.73	2.50	0.47	0.30
Expert-written questions	5	81.66	4.42		

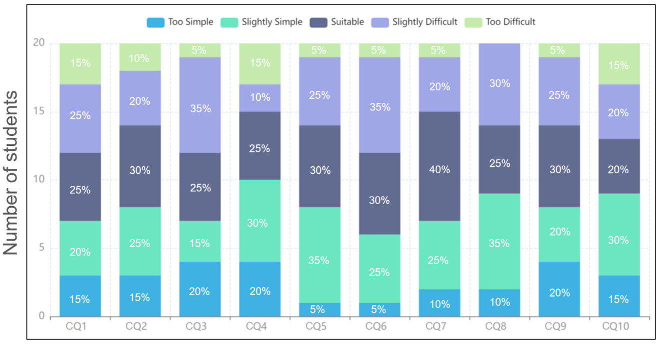

Fig. 9. Adaptability survey on the control group's question difficulty levels.

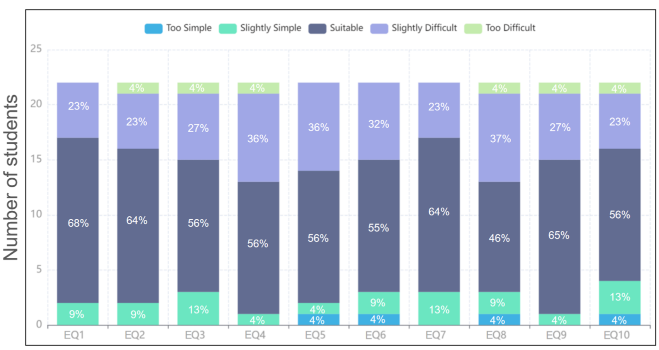

Fig. 10. Adaptability survey on the experimental group's question difficulty levels.

2) Examining Question Difficulty Adaptation: Furthermore, this study examined the suitability of the difficulty level of the machine-generated questions. Specifically, whether the difficulty of the machine-generated questions aligned with the students' present English reading comprehension skills. The results of the difficulty suitability survey conducted on the control group are shown in Fig. 9. It can be seen that due to the control group using standardized questions, the difficulty of the questions seldom corresponded well with the current reading comprehension skills of most students. A substantial portion of students chose responses, such as "too simple," "slightly simple," "slightly difficult," and "too difficult."

In the experimental group, where the students engaged with personalized questions, the question difficulty was adjusted to align with their current reading comprehension skills, as shown in Fig. 10. The survey on the difficulty suitability for the experimental group indicated that the majority of respondents chose

TABLE VII
RESULTS OF ANCOVA ON STUDENTS' LEARNING ACHIEVEMENT AND LEARNING MOTIVATION

Variable	Group	Mean	SD	Adjusted mean	Std. error	F	η^2
Learning achievement	EG	82.36	5.44	81.69	0.98	64.15***	0.62
	CG	69.50	7.76	70.24	1.02		
Learning motivation	EG	3.68	0.51	3.67	0.10	49.72*	0.56
	CG	2.57	0.45	2.58	0.11		

***p <0.01, *p <0.05, experimental group(EG), control group(CG).

the "suitable" option, accounting for almost more than 50%. This outcome further validates the effectiveness of the QG pattern in this study. However, it is worth noting that a notable percentage of students opted for "slightly difficult." This result can be attributed to the fact that their ZPD was taken into account in the prediction stage of the learners' English reading comprehension skill in Section IV-B, making the questions slightly more difficult than their current skill.

Subsequently, the experts categorized the machine-generated questions into different levels of difficulty. Level 1 questions had answers that can be found directly in the article. Level 2 questions require students to understand the corresponding section of the article. Level 3 questions demanded inferences from the context. Level 4 questions called for summarization and provision of relevant examples from the content. Level 5 questions expected students to understand the underlying metaphors and provide their individual perspectives. Remarkably, the experts' rationale for categorizing the questions based on different levels of difficulties is consistent with that of the ChatGPT depicted in Fig. 4.

C. Does the Use of the ChatPRCS System for English Reading Comprehension Learning Result in Increased Levels of Students' Learning Achievement, Learning Motivation, and Cognitive Load?

1) Results of Analysis of Covariance (ANCOVA) on Students' Learning Achievement and Learning Motivation: To verify the differences in learning outcomes and motivation between the experimental group and the control group, Table VII presents the results of the covariance analysis for these two parts.

Having verified nonviolation of the assumption of homogeneity of regression with an $F = 0.071$ ($p > 0.05$), the one-way ANCOVA was conducted by taking the student pretest scores as a covariate. As reported in Table VII, there was a significant difference in learning achievement between the two groups ($F = 64.15$, $p < 0.001$). Adjusted mean scores and standard error were 81.69 and 0.98 for the experimental group and 70.24 and 1.02 for the control group. Notably, the effect size, denoted as eta squared (η^2), was 0.62, indicating a large effect size [55]. This result implied that students engaging with the personalized questions generated by the system performed better in learning achievement.

The study employed a learning motivation questionnaire to discern the motivation differences between the two groups. With $F = 0.003$ ($p > 0.05$), verifying the nonviolation of regression homogeneity assumption, the one-way ANCOVA was conducted. The students' prequestionnaire scores for learning motivation were used as a covariate. As outlined in Table VII, there was a significant difference between the two groups ($F = 49.72$, $p < 0.05$). The eta squared (η^2) reached 0.56, indicating a substantial effect size [55]. This means that the control and experimental groups exhibited varying motivations attributed to their different learning approaches. Notably, the use of the ChatPRCS system for English reading comprehension learning significantly elevated motivation among students in the experimental group.

TABLE VIII
RESULTS OF t-TEST ON STUDENTS' COGNITIVE LOAD

Variable	Group	Mean	SD	t	d
Mental load	EG	3.18	0.60	2.32*	0.71
	CG	2.74	0.64		
Mental effort	EG	2.85	0.79	0.35	0.11
	CG	2.76	0.70		

*p<0.05, experimental group(EG), control group(CG).

TABLE IX
RESULTS OF MANN–WHITNEY U TEST ON STUDENT'S LEARNING ACHIEVEMENT, LEARNING MOTIVATION, AND COGNITIVE LOAD

Variable	Group	Mean rank	Sum of rank	Mann–Whitney U	Asymp.Sig. (2-tailed)
Learning achievement	EG	29.05	639.00	54.00	<0.001
	CG	13.20	264.00		
Learning motivation	EG	30.43	669.50	23.50	<0.001
	CG	11.68	233.50		
Mental load	EG	25.64	564.00	129.00	0.021
	CG	16.95	339.00		
Mental effort	EG	21.70	477.50	215.50	0.909
	CG	21.28	425.50		

Experimental group(EG), control group(CG).

2) Results of t-Test on Cognitive Load: A t-test was conducted to assess the cognitive load experienced by both groups throughout the learning process. As illustrated in Table VIII, the means and standard error of mental load were calculated as 3.18 and 0.60 for the experimental group and 2.74 and 0.64 for the control group. The t-test result showed a significant difference in mental load between the two groups ($t = 2.32$, $p < 0.05$). Specifically, the experimental group demonstrated a significantly higher mental load than the control group. On the other hand, the means and standard error of mental load were computed as 2.85 and 0.79 for the experimental group and 2.76 and 0.70 for the control group. The t-test result, however, showed no significant difference in mental effort between the two groups.

3) Results of Mann–Whitney U Test on Student's Learning Achievement, Learning Motivation, and Cognitive Load: To avoid the impact of small sample sizes, a Mann–Whitney U test was carried out according to the four variables: learning achievement, learning motivation, and cognitive load (including mental load and mental effort) in the learning process. As shown in Table IX, there were statistically significant differences in learning achievement ($p < 0.001$), learning motivation ($p < 0.001$), and mental load ($p = 0.021 < 0.05$). There was no significant difference in mental effort. The results show that despite the different statistical methods used, they still yielded similar results to the conclusions obtained from the initial parametric statistical methods. This is further evidence of the consistency and reliability of our findings.

VIII. Discussion

In this study, the ChatPRCS system was developed for personalized English reading comprehension learning. The quality of machine-generated questions and their influence on students' learning achievement, learning motivation, and cognitive load were explored.

First, based on the prediction results of the selected machine learning models, MLR, DT, and NN demonstrated similar predictive performance. However, considering the inherent characteristics of models, both MLR and DT showed a superior ability to rank the importance of features. Furthermore, in terms of interpretability within an educational context, these models were significantly better than NN. However, this distinction might arise from the interplay between the volume of experimental training samples and the number of features. The prediction performance of NN did not exhibit significant improvement, which was consistent with the findings in [56]. It could also be inferred that NN was likely to achieve better prediction accuracy with a sufficiently large training set.

During the model training process, this study used the cost-sensitive learning paradigm to train the prediction model, which involved setting the ZPD weight matrix. According to (7), this matrix was determined by the interval of ZPD parameter ($\varphi = 1$ in this study). In the specific experimental process, we attempted to set the φ higher in the early stages, such as $\varphi = 1.5$ or 2. By asking some learners about their experience with ChatPRCS to investigate their views on system functionality, we found that when the value of φ was higher, they believed that the system could provide exercises of appropriate difficulty more quickly. But if we continue to follow $\varphi = 2$ in the later stage, most learners reported that the difficulty of the questions exceeded their expectations. This may be due to insufficient data collection on the historical performance of learners in the early stages, and the system's judgment may be biased toward lower level output results. Finally, based on feedback from learners in the later stages of the experiment, we determined $\varphi = 1$. That is, the adjustment process of this value is mainly carried out by teachers based on student feedback. As the amount of data increases in the future, we plan to dynamically adjust this parameter through algorithms based on the overall performance of learners to adapt to their skills.

In evaluating the quality of machine-generated problems, interview findings indicated that most students believed the machine-generated questions provided in this study were roughly consistent with expert-written questions in terms of design and quality. Hence, when categorizing questions, they subconsciously perceived that machine-generated questions might not meet high standards of quality, leading them to often classify them as expert-written questions. This explains why students exhibit a high accuracy rate in discriminating the questions authored by experts and a low accuracy rate when dealing with machine-generated questions. In addition, within the experimental group, a significant percentage of students chose the "slightly difficult" category in the survey assessing the suitability of question difficulty. This result may be attributed to the consideration of the learners' ZPD during the prediction stage of their English reading comprehension skills. In essence, this involves setting the difficulty of some questions slightly above the student's current skills, resulting in the generation of more challenging questions.

In terms of learning achievement, the experimental group was significantly better than the control group. This finding is consistent with prior studies indicating that personalized learning aided by intelligent technology bolsters students' skills. This finding also supports the hypothesis that personalized QG aligned with learners' ZPD enhances learning achievement more effectively, which is consistent with the conclusions from [6], [57]. Moreover, regarding learning motivation, the experimental group outperformed the control group significantly. This result may be due to the ChatPRCS system's adeptness at catering to individual students' needs by generating questions aligned with their reading comprehension skills. ChatPRCS sustains their engagement with the questions, preventing potential frustration and discouragement caused by the questions being excessively challenging or overly simple. In addition, the ChatPRCS system offers timely feedback, including the automated evaluation function that informs students about the accuracy of their responses, enabling them to rectify errors quickly. Several studies have also demonstrated the importance and effectiveness of providing students with immediate feedback to improve their learning achievement and motivation [58].

Concerning cognitive load, there was no significant difference in the mental effort scores between the two groups. This observation may be because the participants majoring in educational technology are familiar with extensive online learning experience. Their interactions with the ChatPRCS system did not cause significant external load. However, the experimental group did exhibit a higher level of mental load than the control group. This finding corresponds to the dynamic difficulty threshold incorporated during the personalized QG phase, affirming the efficacy of this design. The marginally elevated difficulty level of personalized questions aligned more harmoniously with their respective ZPD, culminating in superior learning achievements (corresponding to the two groups' previously obtained learning achievement). This is also evident in the question difficulty survey presented in Fig. 10, providing further support for the aforementioned findings.

However, this study still has some limitations. Due to the data volume constraints, the current assessment of learners' reading comprehension skills relies heavily on the available experimental data. In terms of evaluating the quality of generated questions, this study primarily relies on subjective learner assessments. In the future, an evaluation method that combines both manual and automatic evaluations should be developed to improve the quality of personalized QG. Moreover, attention must be paid to the ethical concerns arising from generative models. Safeguarding student privacy and ensuring data security during personalized question design remain of utmost importance. Steps must be taken to prevent generative models from inducing bias or discrimination, ensuring fairness and impartial outcomes.

IX. Conclusion

In this article, a system named ChatPRCS was designed for English reading comprehension learning, and a reading comprehension skill prediction algorithm and specific prompt mode are proposed for ChatPRCS. Building upon ChatGPT and the ZPD theory, the ChatPRCS system can adapt to individual student skills, further enhanced by the incorporation of prompt patterns. A quasi-experimental study was conducted to evaluated the effect of ChatPRCS on students' learning achievement, motivation, and cognitive load. The results demonstrate the capacity of the ChatPRCS system to generate personalized, high quality, distinctive reading comprehension questions. More specifically, ChatPRCS enhances learners' reading comprehension skills and boosts their learning motivation, while minimally increasing their cognitive load.

The future work of this study will include the following:

1) to capture students' needs more accurately, explore techniques to incorporate a greater volume and diversity of data into skill prediction and QG models;
2) to enhance transparency and controllability in the QG process, delve into the interpretability of generation models in conjunction with logic structures;
3) to improve the effectiveness of personalized QG, integrate other effective and advanced technologies into generation models.

These advanced technologies are expected to be more seamlessly integrated with the education system, playing a pivotal role in improving students' reading comprehension skills.

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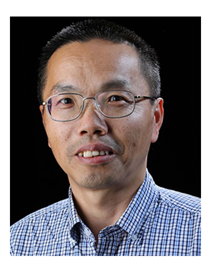

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